DATA-DRIVEN PREDICTION AND ANALYSIS OF COOLING ENERGY USE IN TELECOM CENTRAL OFFICE ENVIRONMENTS

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Outline

- Research motivation and gap in telecom cooling analytics
- Target engineering from fragmented operational telemetry
- Forecasting methodology and expected-demand benchmarking
- Forecast-informed supervisory control simulation and results
- Conclusions, contribution, novelty

1. Problem and motivation

- Telecom central offices require **continuous cooling for reliable operation of sensitive equipment**
- Cooling is a **major infrastructure energy consumer**, so its performance matters for site-level efficiency management
- Existing telecom analytics are stronger in **monitoring and diagnosis** than in transforming fragmented telemetry for **forecast-based operational decision support**

2. Research gap and literature background

- Telecom studies are relatively strong in **monitoring, KPI design, and diagnosis**
- Real cooling telemetry is often **not analysis-ready** and may require **target reconstruction** from fragmented cumulative measurements
- Forecasting and control literature provide useful methods, but often assume **ready-made targets** or **richer control data** than are available here

Research gap: no integrated telecom-focused framework from **target engineering** to **forecasting, benchmarking, and supervisory control** under real telemetry constraints

Sorrentino et al. (2024); D'Aniello et al. (2018); Malafronte et al. (2021); Fan et al. (2021); Ji et al. (2017);

3. Research aim and objectives

Aim:

To develop predictive models for cooling energy use and evaluate intelligent control strategies based on real monitoring data from telecom central office environments.

Objectives:

1. conduct the literature review
2. engineer an hourly cooling-demand target and analytical dataset
3. develop and compare short-term forecasting models
4. analyse actual versus expected cooling demand
5. design and evaluate forecast-informed intelligent control strategies

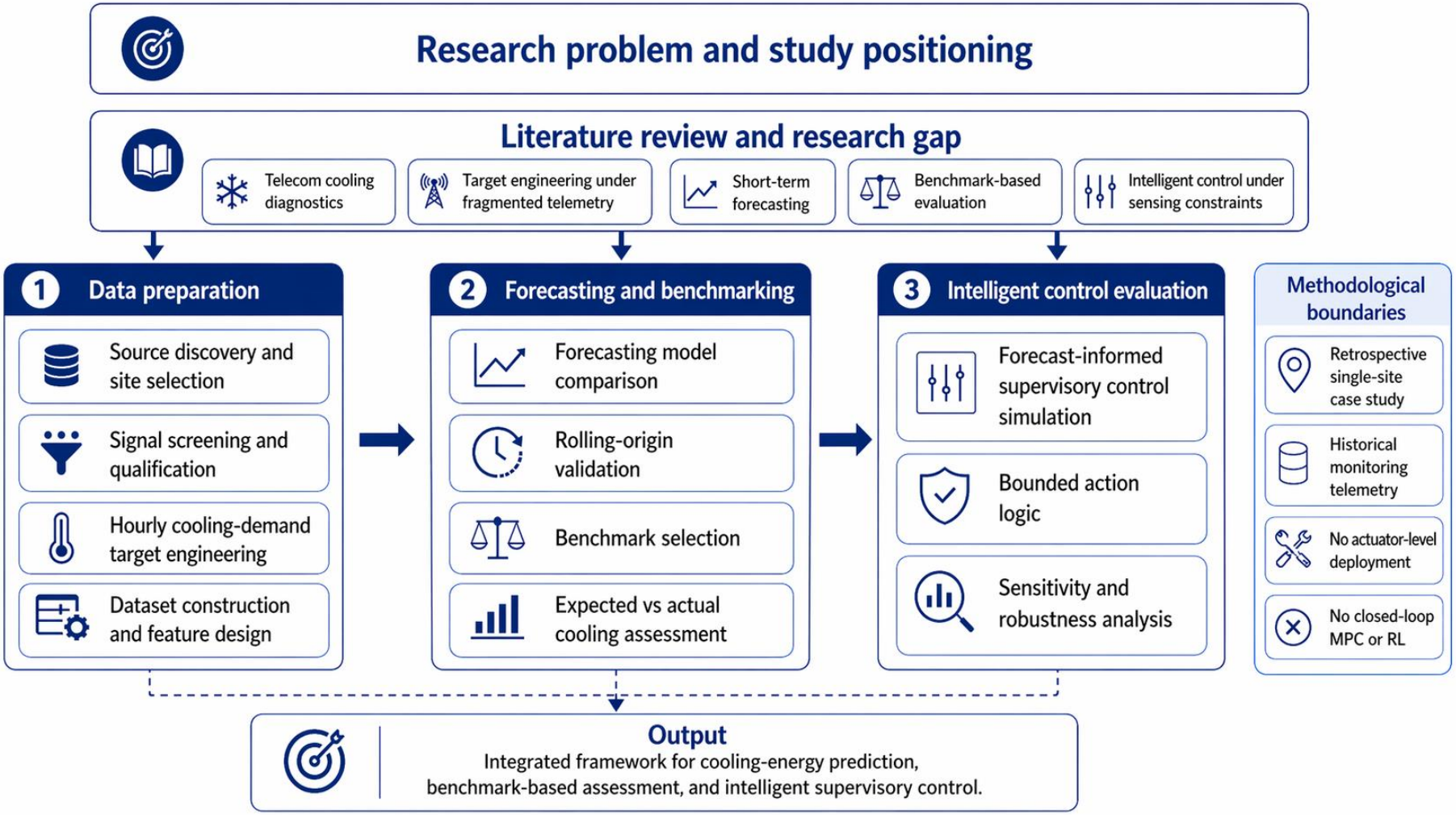
4. Research questions

1. How can real operational and environmental data be transformed into a defensible hourly cooling-demand target and analytical dataset?
2. Which forecasting approach provides the most reliable short-term benchmark for expected cooling demand?
3. To what extent can forecast-informed intelligent control strategies identify conservative opportunities to improve cooling energy use?

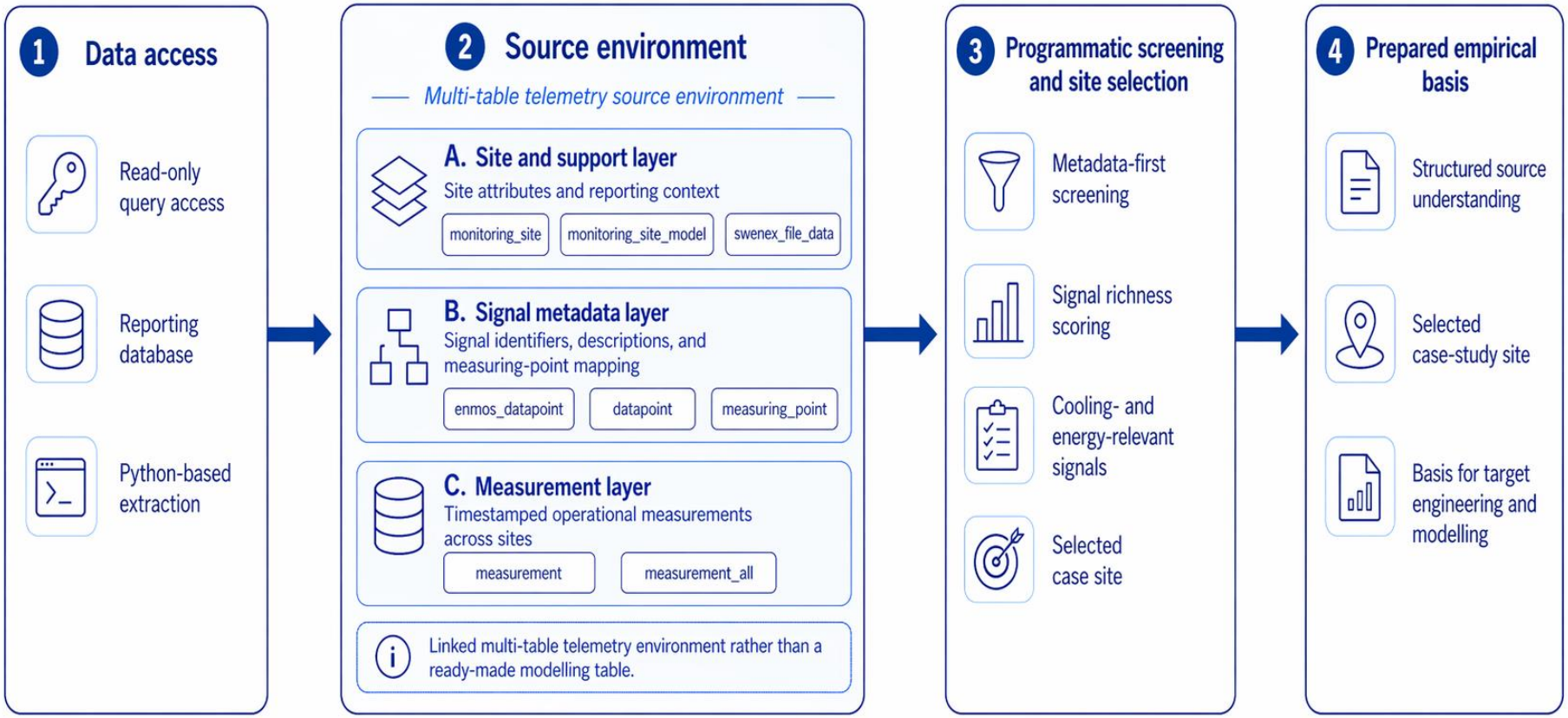
5. Object and subject of research

- **The object of the research** is cooling energy use in telecom central office technical infrastructure.
- **The subject of the research** is data-driven prediction, benchmark-based assessment, and intelligent control strategy simulation for cooling energy use in telecom central office environments.

6. Analytical framework

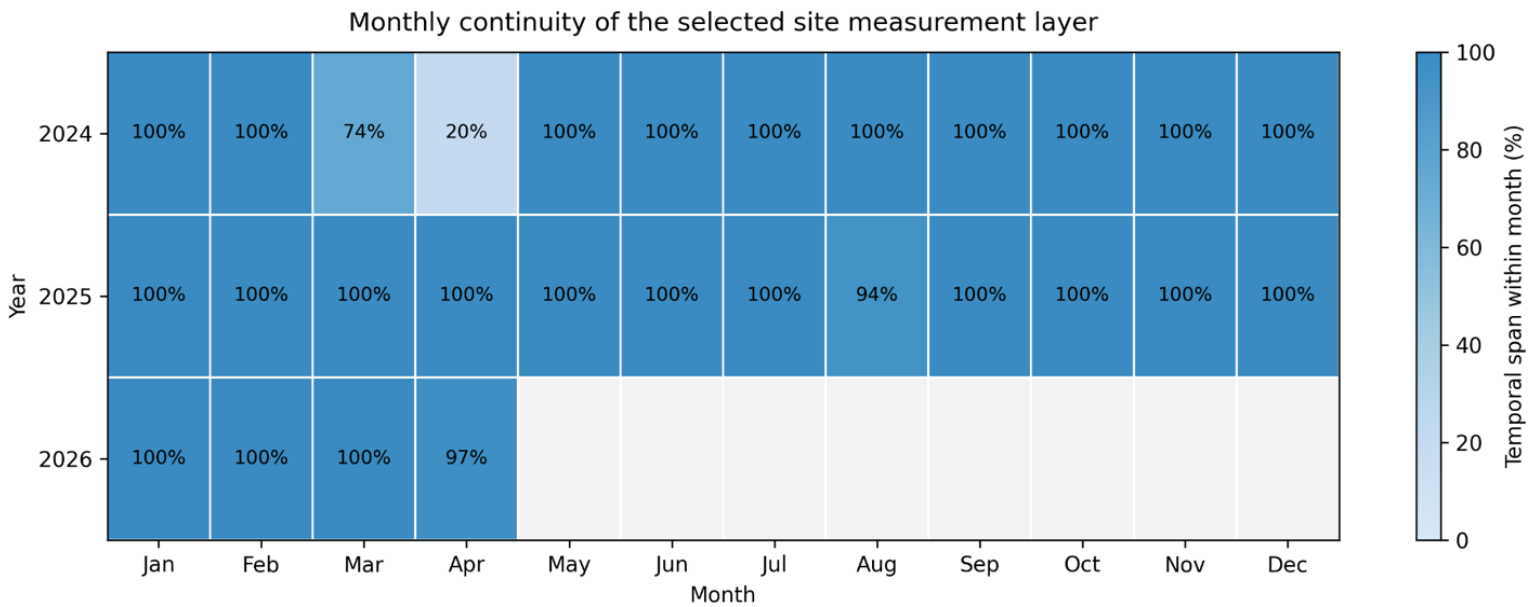


7. Data and case-study context: Data access and source environment



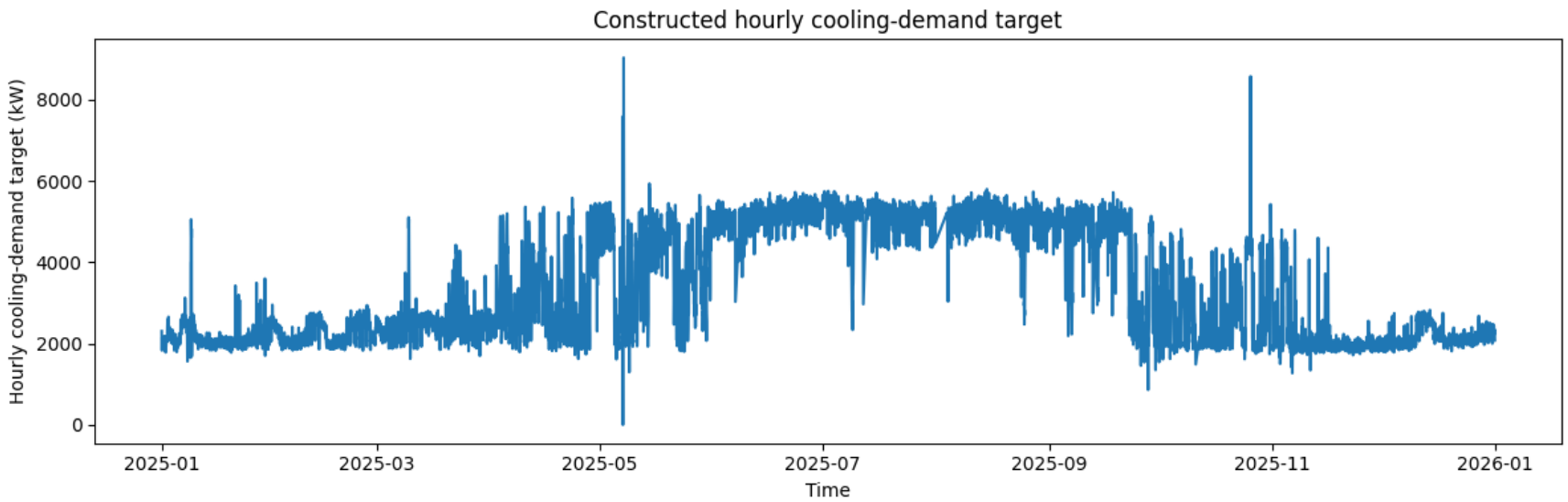
7. Data and case-study context: Study period and temporal continuity

- The data covered more than one year, but the analysis was restricted to **year 2025**
- 2025 is **complete and internally coherent** annual window
- Monthly continuity in 2025 remained effectively complete, only one dip to **94% in August**



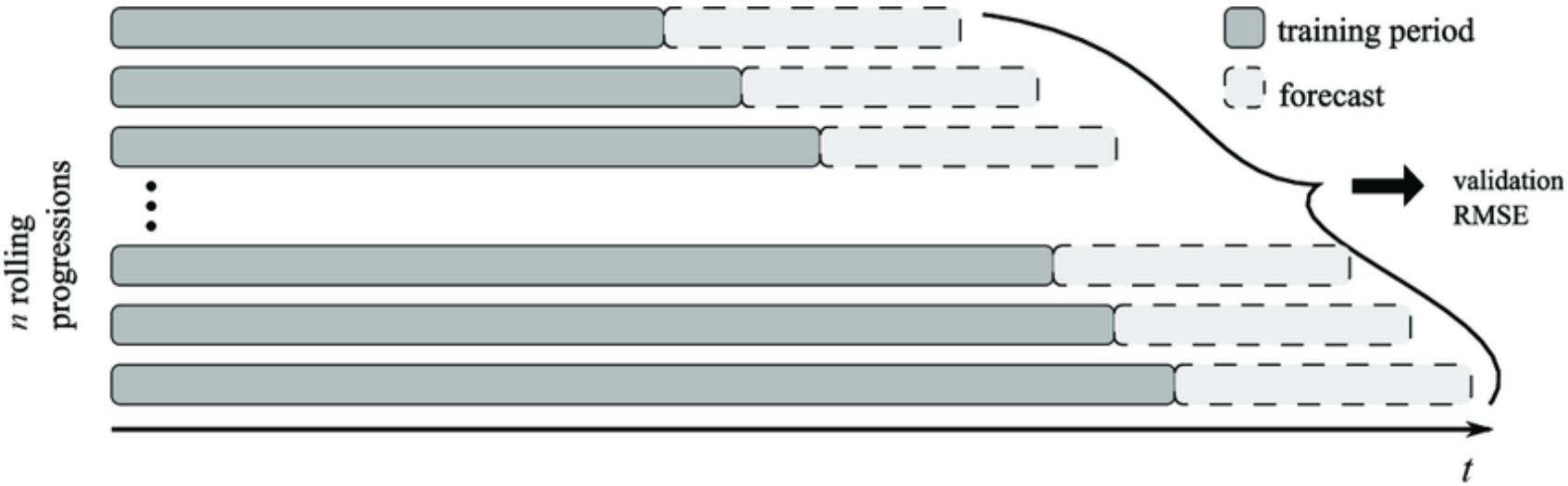
8. Methodology: Target engineering

- Cooling demand was **not available as a ready hourly target**
- Candidate cumulative cooling signals were screened
- Hourly target built by alignment, differencing, clipping, and aggregation



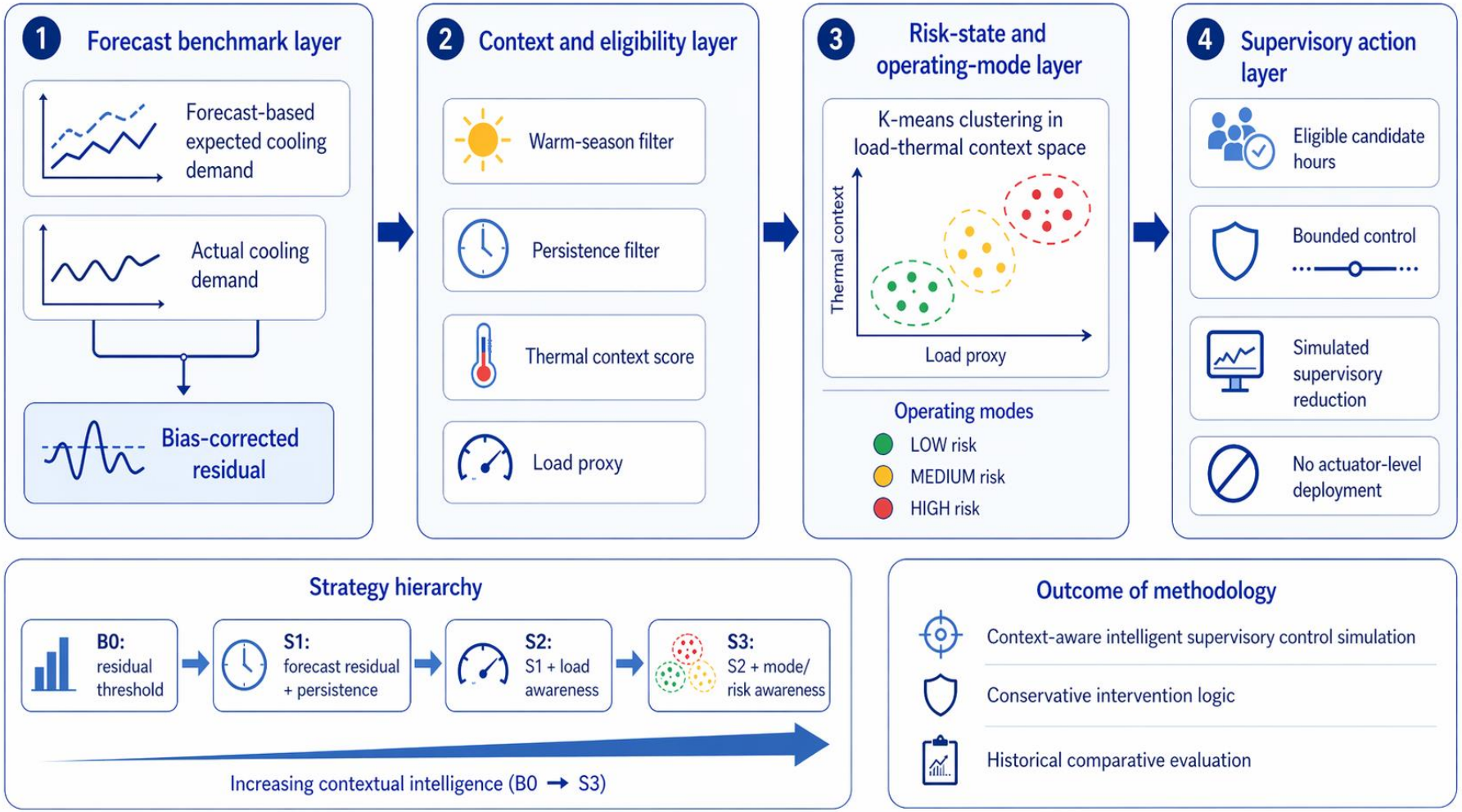
8. Methodology: Forecasting and validation

- Models compared: naive, linear regression, RF, GB, SARIMAX, LSTM
- Final lag set: **1, 2, 3, 6, 12, 24, 48, 168 hours**
- Metrics: **MAE, RMSE, R^2**
- Validation: **chronological split + rolling-origin**



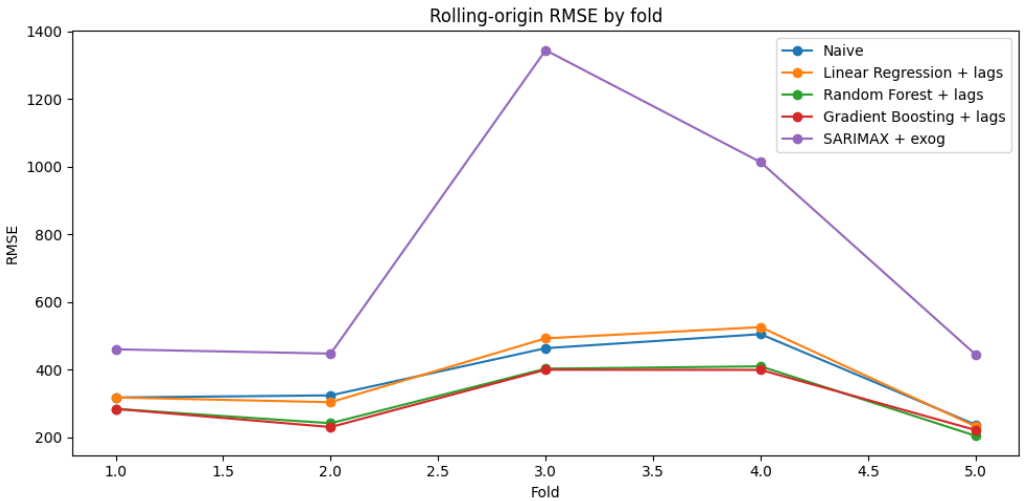
Rolling origin forecast cross-validation principle. Figure adopted from Oliveira and Ramos (2019)

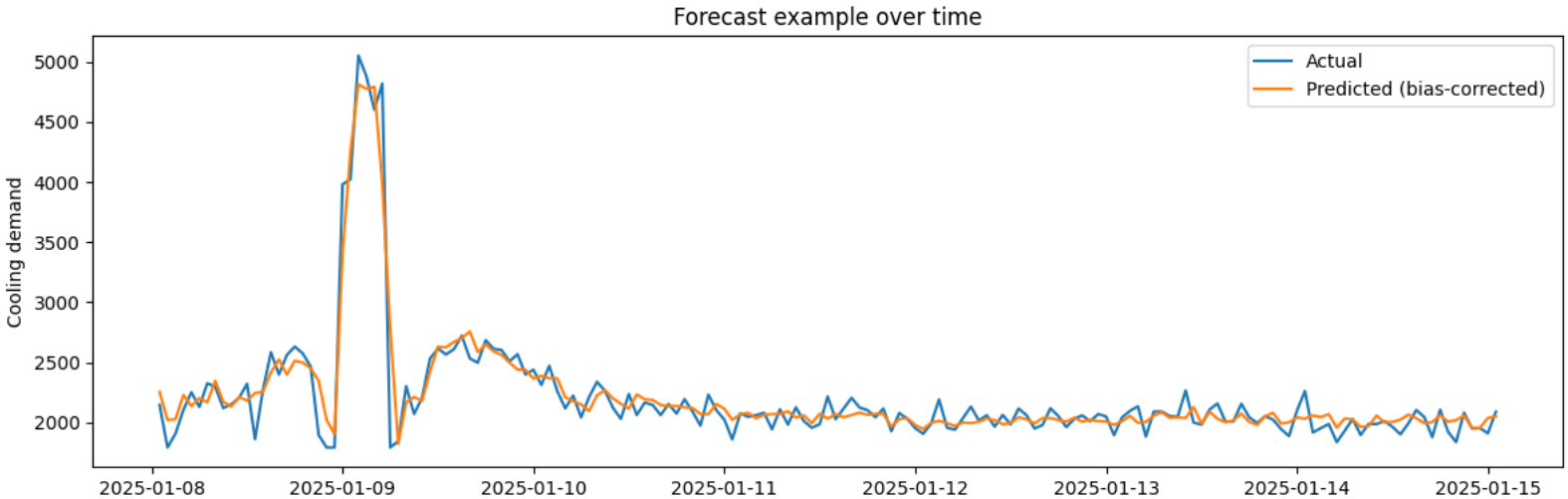
8. Methodology: Forecast-informed intelligent control strategy



9. Results: Forecasting performance and final model selection

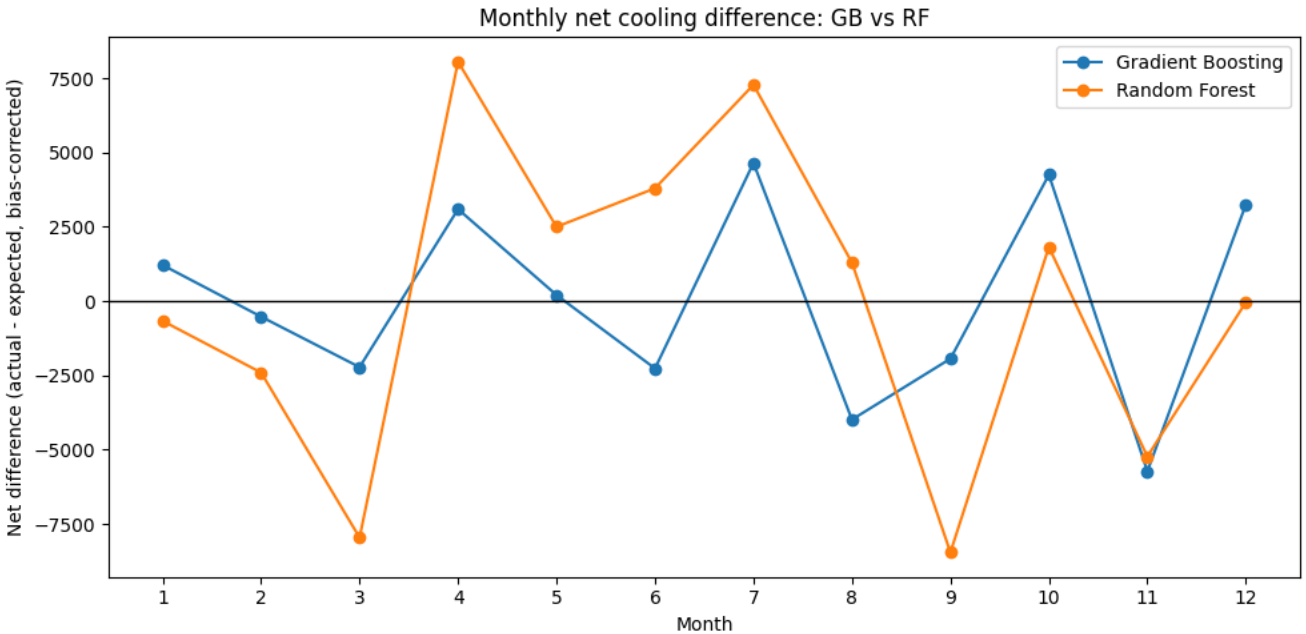
- Best model: **Gradient Boosting + lags**
- Strong secondary benchmark: **Random Forest + lags**
- Final benchmark quality: **MAE 115.06, RMSE 161.79, R^2 0.9855**





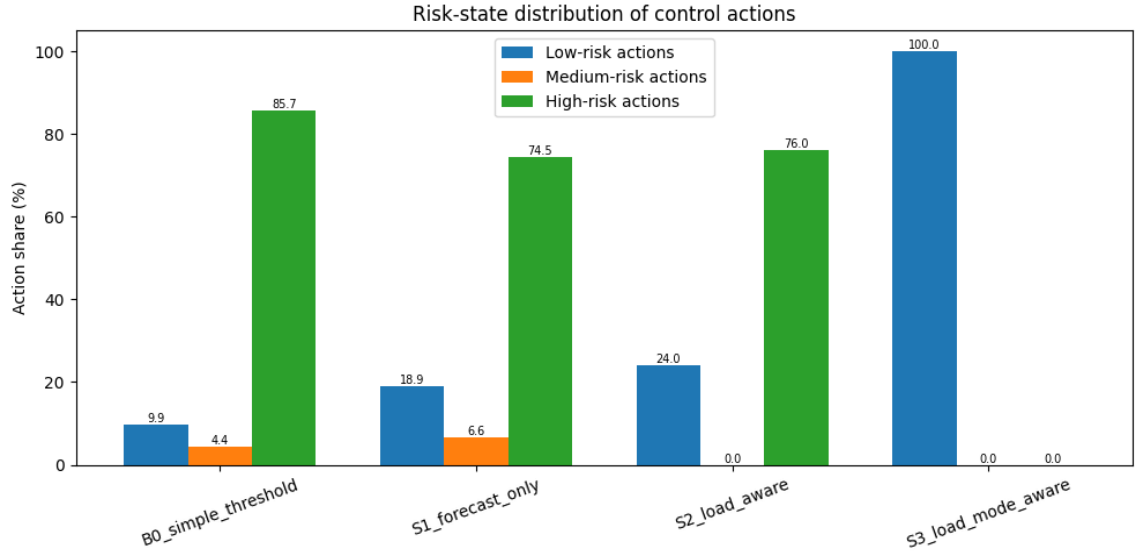
9. Results: Benchmarking actual vs expected cooling

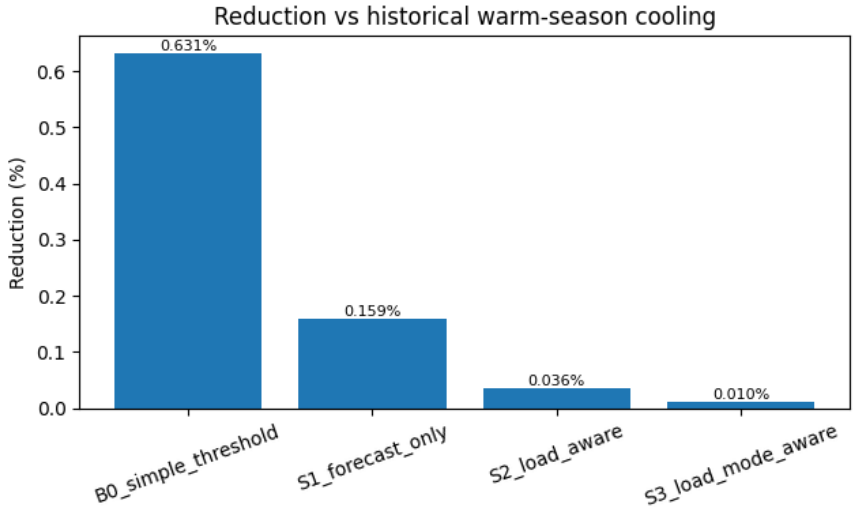
- Full-year **net difference ≈ 0** after bias correction
- Warm season: **excess 1.34%, reduced 1.36%, net -0.02%**
- No evidence of large-scale systematic annual overcooling

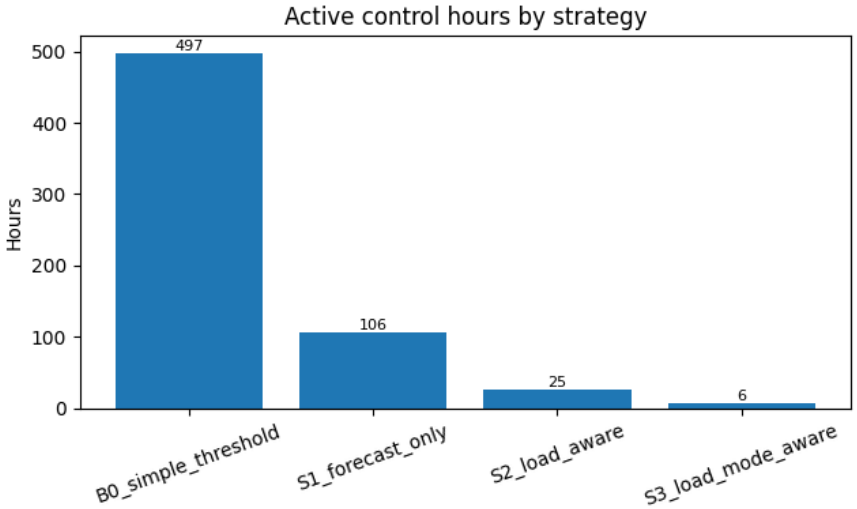


9. Results: Control outcome

- Increasing supervisory context yields **fewer active hours and smaller reduction estimates**
- Stricter strategies also shift actions toward **safer operating**

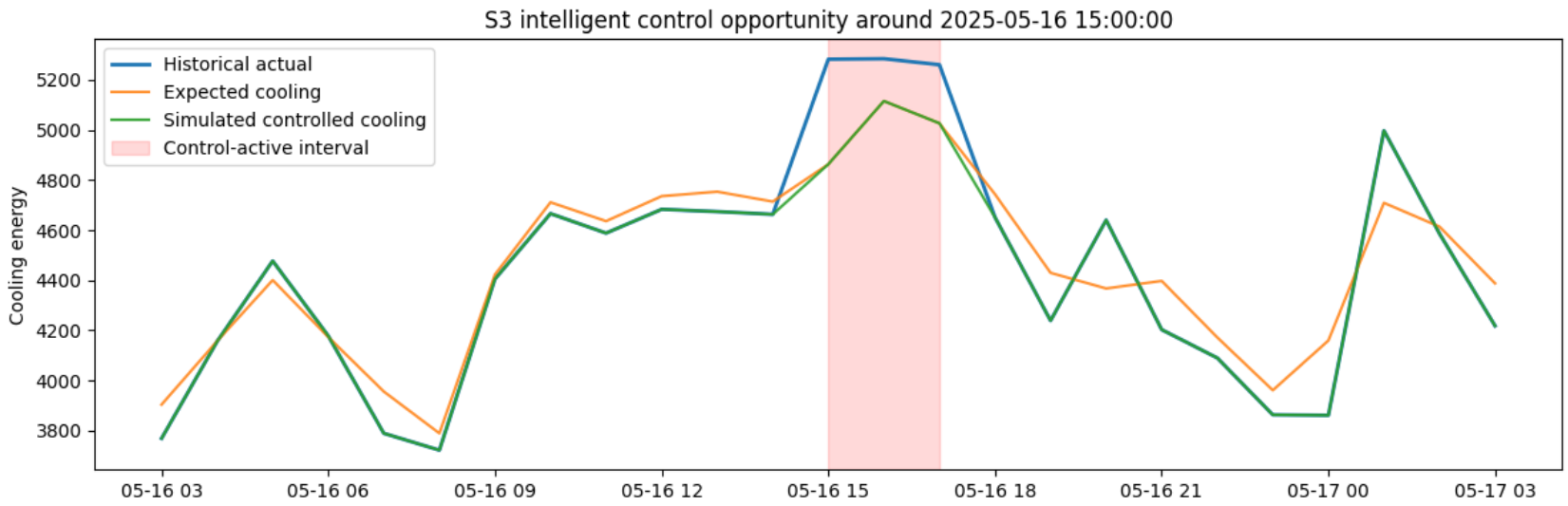






9. Results: Example of simulated control intervention

- Actual cooling exceeded the **expected-demand benchmark**
- Simulated control reduced cooling only during the **eligible interval**
- Intervention remained **conservative and time-local**



10. Limitations

1. **Single-site case study** - the study prioritised methodological depth and traceability under one rich telemetry environment rather than broader but shallower cross-site comparison
2. **Engineered target rather than directly observed demand** - cooling demand was not available as a ready hourly variable and had to be reconstructed from fragmented cumulative subsystem telemetry.
3. **No deployed closed-loop control** - the available data did not provide the manipulated variables, validated process model, or safe control environment required for full MPC or RL.
4. **Control cannot be validated by exact accuracy** there is no ground-truth optimal action sequence, so control trust is defended through effectiveness, selectivity, risk-awareness, and robustness.

Conclusions

- The study **successfully reconstructed a defensible hourly cooling-demand target** from fragmented cumulative subsystem telemetry by retaining signals **120, 230, 46, 49, and 78** and converting them into an engineered hourly analytical series.
- Under the final rolling-origin validation stage, **Gradient Boosting with lagged target features** was selected as the primary benchmark model, achieving **RMSE_pooled = 316.64** versus **319.49** for Random Forest, and final one-hour-ahead quality of **MAE = 115.06**, **RMSE = 161.79**, and **R^2 = 0.9855**.
- Benchmark-based assessment showed **no evidence of large-scale systematic annual overcooling**: for the selected Gradient Boosting benchmark, warm-season **excess = 1.3407%**, **reduced = 1.3610%**, and **net = -0.0203%**; Random Forest produced a closely aligned result.

Conclusions

- Forecast-informed supervisory control nevertheless identified a **small, bounded, and context-aware intervention margin**: the warm-season candidate set narrowed from **3537** hours to **1832** positive-residual hours, **458** meaningful-excess hours, and **38** control-eligible hours; final strategy results were **S1: 106 h / 0.159%**, **S2: 25 h / 0.0356%**, and **S3: 6 h / 0.0102%**.
- The main contribution of the thesis is an integrated telecom cooling analytics framework linking target reconstruction, short-horizon forecasting, benchmark-based assessment, and supervisory control simulation under real telemetry constraints.

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Thank you, any questions?

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